Response to the editor and reviewers

Response to the editor and reviewers. Reviewer 1 denotes the downloaded report (10 major and 7 minor comments); Reviewer 2 reported inaccessible figures; Reviewer 3 denotes the six comments supplied in correspondence, following the journal numbering confirmed by the corresponding author. Original comment text is reproduced below. Page references identify the revised PDFs supplied with this response. The revision incorporates subsequent human author review and a separate descriptive GO analysis, with their timing and scope stated explicitly.

We thank the editor and reviewers for helping distinguish transparent hypothesis generation from validated disease-gene prediction. We have separated seed-free raw scoring from control-informed HIGH calibration, reported final control tiers and filtering losses, added matched disease comparators and simple baselines, quantified weight/completeness/proxy dependence, and strengthened numerical replay and delivered-file checksums. We also completed a retrospective temporal assessment with 45 later target associations fixed before new ranking calculation. Its weak recovery is reported directly and narrows the claims. Following the initial AI-assisted ascertainment, human author review examined primary reports for all 45 cases, including reported families/pedigrees, phenotypes, inheritance, functional mechanisms and publication chronology. The review highlighted distinct disease mechanisms and family-evidence concerns involving PDIA6, VWA8 and PLCG1, detailed in R1-major-8. Production scores, weights and thresholds were not optimized against these new outcomes.

Editor

Comment.

> Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained.

Response. We have revised the interpretation to a transparent framework for hypothesis generation, rather than established Usher-specific prediction or reliable novel disease-gene discovery. The revised analyses report the control-calibration dependence of HIGH, the high raw ranks of housekeeping genes, sensitivity of candidate shortlists to weighting and expression proxies, and the weak retrospective temporal recovery. HIGH is described as a heuristic priority tier, not a calibrated disease probability or independently validated measure of specificity. The additional author review described in R1-major-8 and R3-4 provides a clinical and biological check on the temporal ascertainment, while retaining unresolved eligibility concerns explicitly. We do not treat this review as a blinded prospective validation or claim that it improves the observed prediction results.

Reviewer 1 — major comments

R1-major-1. Define the prediction task; separate Usher and SYSCILIA controls

Comment.

> The intended prediction task is not entirely clear
>
> This paper describes UsherPipe as a gene prioritization tool related to Usher syndrome in several parts, but in others, it extends its scope to Usher syndrome and related ciliopathy. These are related, but they are not the same prediction problem.
>
> This distinction matters because the evaluation set includes nine established Usher genes together with 28 selected SYSCILIA genes. Good recovery of general ciliary genes does not necessarily demonstrate good performance for Usher-specific gene discovery.

I would like the authors to define the main objective more clearly. Does this pipeline aim to identify new genes associated with Usher syndrome, genes associated with sensory ciliary disease, or a broader range of cilia-related genes that could be studied further in relation to Usher syndrome?

The results for the nine Usher genes and the 28 SYSCILIA genes should also be reported separately throughout the manuscript rather than relying mainly on the combined set of 37 controls.

Response. The primary task is genome-wide cilia/sensory-aware hypothesis generation for follow-up in the Usher context. It is neither an Usher-specific diagnostic classifier nor proof that broad ciliary genes cause the combined auditory–retinal phenotype. The Abstract, Background, Results and Conclusions now distinguish the nine established Usher controls from the 28 selected SYSCILIA controls. Raw median percentiles are 97.77% and 91.93%; final HIGH counts are two and one. The curated SYSCILIA subset is not a random validation sample. Pooled counts are retained only where describing the historical cached benchmark or original combined artifacts, with the group composition explicit.

The temporal cohort likewise covers a broader, prespecified target comprising inherited retinal/optic disease, hearing impairment and explicitly supported ciliopathies. Human author review reinforced that a sensory phenotype alone does not establish a ciliary mechanism: mitochondrial, retinal vascular, glycosylation and RNA-splicing disorders are represented among the nominated cases. These cases cannot collectively be described as newly identified ciliopathy or Usher genes. Mechanistic examples and the review process are given in R1-major-8; functional classification and the clinical family-evidence criterion are assessed separately.

Changes/evidence. Main Background “Usher syndrome as a specialized sensory ciliopathy”, Results “Internal control recovery” and Table 3; results/control_rankings.tsv, results/control_tier_trace.tsv; Supplementary Methods S4.

Revised PDF locations. Manuscript p. 1–2 (Abstract); Manuscript p. 3–4 (Usher syndrome as a specialized sensory ciliopathy); Manuscript p. 19–20 (Internal control recovery); Manuscript p. 32 (Conclusions); Supplementary Methods p. 4 (S4. Control traces).

R1-major-2. Calibration and HIGH evaluation reuse the controls

Comment.

The calibration and evaluation of the HIGH tier are not independent

A central concern is that the cilia-signal gate was chosen after examining the behavior of both positive and negative controls.

The authors are commendably transparent about this, but it creates an important circularity. Once the controls have been used to inform the definition of the gate, those same controls cannot provide an independent assessment of how well the final HIGH tier performs.

This is particularly relevant because the HIGH tier is one of the main outputs of the pipeline.

Ideally, the authors should separate calibration and evaluation using an independent set of genes. This could be done using a held-out control set, a temporal split, or another pre-specified development/test design.

If such an analysis is not feasible, I would recommend describing the HIGH tier consistently as a heuristically calibrated prioritization tier, rather than implying that its specificity has been validated.

The distinction between the raw six-layer score and the post-hoc gated HIGH tier should also be made more prominent in the Abstract and Results.

Response. We agree. The downstream gate was selected after inspection of positive/housekeeping behavior, although the raw mean uses no disease training labels or similarity seeds. The Abstract and Methods now identify HIGH as a heuristically calibrated priority tier. We corrected the gate description to the actual direct-

compartment/compendium source flags OR the positive-animal Q75 threshold, not any positive aggregate localization score. Internal control recovery and removal of housekeeping genes are calibration diagnostics, not independent specificity validation. Moreover, 26/28 SYSCILIA controls occur in the embedded compendium that feeds localization scoring and gate passage; no Usher control occurs there. This creates partly circular ciliary-control recovery even without training labels.

We added a separate later-association cohort selected from clinical/date evidence before new rank calculation. Two co-primary historical analogues evaluated the 41 protein-coding cases mapped to the historical universe; four noncoding cases remained failures in the original 45-case end-to-end denominator. These denominators describe the frozen AI-assisted ascertainment and have not been replaced by a newly finalized human-adjudicated cohort. Subsequent human review and unresolved family-evidence concerns are described in R1-major-8. Neither version recovered a principal case at top 100 or HIGH. Historical genome-wide HIGH counts were only zero/five, limiting that endpoint's information. All 41 had at least four observed weighted families, but none met score 0.7, so gate failure alone cannot explain the result. The control-informed gate architecture and post-cutoff method development remain disclosed; the temporal assessment does not establish clinical specificity or prospective validity. We consistently describe HIGH as a heuristically calibrated priority tier.

Changes/evidence. Abstract; Methods "Retrospective evaluation of historical analogues"; Results "Retrospective temporal evaluation" and Table 4; Discussion "Limitations"; S6; Additional file 12, including original/amended pre-outcome manifests and all 23 scheme outcomes.

Revised PDF locations. Manuscript p. 1–2 (Abstract); Manuscript p. 11–12 (NULL-aware composite scoring); Manuscript p. 12–14 (Retrospective evaluation of historical analogues); Manuscript p. 24–26 (Retrospective temporal evaluation); Manuscript p. 29–31 (Limitations); Supplementary Methods p. 2–3 (S3. Implemented); Supplementary Methods p. 5–11 (S6. Retrospective).

R1-major-3. Show final control tiers, reasons for losses and threshold sensitivity

Comment.

> Recovery of known genes in the final HIGH tier needs to be shown directly
>
> The manuscript emphasizes that 35 of 37 controls fall within the top quartile of the genome-wide ranking. This sounds encouraging, but it does not tell the reader how the known genes behave after the complete prioritization procedure is applied.
>
> According to the supplementary analyses, only two established Usher genes are present among the 62 HIGH-tier genes, and only three controls in total are retained when the selected SYSCILIA genes are also considered.
>
> For a pipeline whose practical output is a relatively small HIGH-priority shortlist, this is an important result and should be presented clearly in the main manuscript.
>
> I suggest reporting:
>
> how many of the nine known Usher genes fall into HIGH, MEDIUM, LOW, or the excluded group;
>
> the same information for the SYSCILIA controls;
>
> why each established Usher gene that fails to reach HIGH is lost—i.e., score threshold, evidence count, or cilia-signal gate;
>
> how the sensitivity of the HIGH tier changes if alternative thresholds are used.
>
> A simple table tracing the known Usher genes through the different filtering steps would be very helpful.

Response. The main Results now report Usher HIGH/MEDIUM/LOW/excluded = 2/7/0/0 and SYSCILIA = 1/27/0/0. Table 3 traces every Usher control. All seven non-HIGH Usher controls fail composite ≥0.70; WHRN and USH1G additionally fail the gate. None fails the layer-count criterion. With the gate and minimum three layers fixed, thresholds 0.60/0.65/0.70/0.75/0.80 yield 366/170/62/11/1 HIGH labels, retaining 5/4/2/0/0 Usher and 6/1/1/0/0 SYSCILIA controls. We report this full family without choosing a more favorable production threshold.

Changes/evidence. Main Table 3 and adjacent Results; results/control_tier_trace.tsv, results/threshold_sensitivity.tsv; S4.

Revised PDF locations. Manuscript p. 19–20 (Internal control recovery); Supplementary Methods p. 4 (S4. Control traces).

R1-major-4. Raw housekeeping specificity failure; larger matched disease comparators

Comment.

> The housekeeping-gene result raises a substantial specificity concern
>
> One of the most notable results is that housekeeping genes rank very high in the base score. The median percentile was reported at 94.8%, and 11 of the 13 housekeeping genes fall within the top 25%.
>
> The authors correctly describe this as a specificity failure. I agree with that interpretation.
>
> However, the subsequent removal of these genes from the HIGH tier is less convincing as evidence of improved specificity because the gate was designed after examining these same negative controls.
>
> I also wonder whether housekeeping genes are the most informative negative set for this task. Many of these genes are highly constrained, extensively annotated, and well studied—features that naturally produce high scores in several of the six layers.
>
> A larger and better-matched negative-control set would strengthen the analysis considerably. For example, the authors could consider well-characterized genes associated with diseases unrelated to auditory, retinal, or ciliary biology, with matching for publication burden or annotation density where possible.
>
> Without such an analysis, I would avoid interpreting the current results as an estimate of biological specificity.

Response. We retain the raw housekeeping median of 94.8% as a specificity failure; the gate does not change raw rankings. We added a comparator analysis whose protocol and 111-pair roster were committed before outcome inspection. A Strong/Definitive ClinGen pool excluded recorded auditory/retinal/ciliary disease and HPO annotations and matched three unique genes per control on standardized log1p publication and GO counts, with a fixed 0.75-SD coordinate caliper and no reuse. The 27 and 84 comparator groups had median raw percentiles 64.21% and 60.30%, versus 97.77% and 91.93% for controls; pair concordances were 24/27 and 72/84.

None of the 111 matches was HIGH, but 110 already failed the raw score threshold, so this cannot be credited to gate specificity. The broader 420-gene pool retained three HIGH genes with residual sensory biology identified in a subsequent source check. They were not removed after outcomes. Coverage differs between groups; additional post-outcome common-observed-layer analyses preserve separation, whereas joint animal/literature removal reduces concordance to 55.6%/57.1%. We explicitly discuss matching only two measured research-burden quantities, shared curation, eligibility–feature overlap, and absence of verified biological negatives. These are descriptive specificity diagnostics, not clinical false-positive rates or novel-gene validation.

Changes/evidence. Results “Matched unrelated-disease comparators”; Discussion; S5; phase3_spec.md, phase3_roster/, phase3_results/, phase4_results/fixed_pair_summary.tsv and matching_completeness_balance.tsv.

Revised PDF locations. Manuscript p. 20–21 (Negative controls); Manuscript p. 21–22 (Matched unrelated-disease comparators); Manuscript p. 29–31 (Limitations); Supplementary Methods p. 4–5 (S5. Frozen).

R1-major-5. Heuristic weights, simple baselines and top-K persistence

Comment.

> The choice of weights remains somewhat arbitrary
>
> The six evidence-layer weights are biologically motivated, but the exact values appear heuristic.
>
> This is not necessarily a problem for a transparent prioritization framework, but the sensitivity analysis shows that the resulting ranking is not uniformly stable. Only 16 of 24 perturbations meet the predefined stability threshold, and some correlations fall to approximately 0.61. The animal-model layer appears to have particularly strong influence.
>
> The post-hoc logistic-regression experiment is also informative in this regard. It places most of the fitted weight on the animal-model layer and produces a substantially different genome-wide ranking.
>
> I agree with the authors that such fitted weights should not replace the default model because the same controls are involved in model fitting. Nevertheless, these findings suggest that the default weights should be presented as one reasonable biological configuration rather than as a particularly robust or optimal solution.
>
> It would strengthen the manuscript to include several simple comparisons, such as:
>
> equal weights;
>
> animal-model evidence alone;
>
> expression alone;
>
> localization/cilia evidence alone;
>
> leave-one-layer-out rankings.
>
> It would also be useful to report how frequently genes remain in the top 25, 50, or 100 across plausible alternative weighting schemes.

Response. The exact weights are heuristic, not optimal. Methods now explains the modest five-percentage-point emphasis on constraint/expression and equal weighting of the other layers, while stating that this does not establish superiority. We added equal, all six individual-layer and all six leave-one-layer-out comparisons. Animal-only gives the strongest Usher recovery (4/9 at top 100), whereas literature-only gives the strongest SYSCILIA recovery (21/28; Usher 2/9), compared with default 2/9 and 1/28. Literature-only medians are 99.13%/99.78%, with all nine Usher controls in its top 10%. Known disease terms and research/curation dependence limit these internal results. We report this unfavorable comparison directly.

Equal weighting gives whole-ranking ρ=0.9952 and 86 shared top 100 genes. Across equal plus 24 modest perturbations, only 8 default top25, 17 top50 and 41 top 100 genes persist in every configuration. The finite-family frequencies are not probabilities. Global correlations (0.9598–0.9994) coexist with meaningful shortlist movement; the earlier mean 0.8548 statistic is conditional on overlap, not whole-ranking stability. Logistic fitting (ρ=0.646 versus default; animal weight 0.903) remains a control-reuse diagnostic, not a production replacement.

Changes/evidence. Methods “NULL-aware composite scoring”; Results “Sensitivity analysis”; S2/S4; results/baseline_comparison.tsv, candidate_membership_stability.tsv, control_rankings.tsv.

Revised PDF locations. Manuscript p. 11–12 (NULL-aware composite scoring); Manuscript p. 23–24 (Sensitivity analysis); Supplementary Methods p. 2 (S2. Data-driven); Supplementary Methods p. 4 (S4. Control traces).

R1-major-6. Sparse observed-layer means and completeness thresholds

Comment.

> NULL-aware scoring has advantages, but it may also favor genes with sparse evidence
>
> I find the explicit treatment of missing values to be one of the more interesting aspects of the pipeline. Treating missing evidence as zero can clearly be problematic.
>
> At the same time, the current available-layer weighted mean introduces another issue. A gene with only a few observed layers can receive a high composite score if those observed values happen to be high.
>
> The evidence-count requirement partially addresses this, but a HIGH-tier gene may still be supported by only three of the six evidence layers.
>
> The manuscript would benefit from a more direct analysis of this issue. For example:
>
> How many HIGH-tier genes have evidence from 3, 4, 5, or 6 layers?
>
> Does the HIGH list change substantially if at least four or five observed layers are required?
>
> Is composite score associated with evidence completeness?
>
> In addition, authors are encouraged to provide clear measures of evidence completeness along with the scores. A composite score of 0.75 based on three factors does not necessarily have the same evidential weight as a score of 0.75 supported by all six factors.

Response. All 62 HIGH labels actually have five/six layers (33/29); none has three/four. Requiring four or five retains all 62, while six retains 29. Nevertheless, eight one-layer genes appear in the raw top 100 and are all LOW. Composite score correlates with observed-layer count (ρ=0.2402; n=20,053). We retain per-gene counts/source flags and explain that available-layer means avoid an automatic missingness penalty but do not equalize evidential strength or quantify uncertainty. The historical audit also exposes sparse high-ranking genes; this limitation is not dismissed because production HIGH is well-covered.

Changes/evidence. Results “Impact of missing data handling”; Discussion; S4/S6; results/evidence_completeness.tsv, threshold_sensitivity.tsv, per_gene_diagnostics.tsv; phase4_results/historical_topk_completeness.tsv.

Revised PDF locations. Manuscript p. 22–23 (Impact of missing data handling); Manuscript p. 23–24 (Sensitivity analysis); Supplementary Methods p. 4 (S4. Control traces); Supplementary Methods p. 5–11 (S6. Retrospective).

R1-major-7. Inter-layer overlap and within-layer combination

Comment.

> The six evidence layers may not be as independent as the framework implies
>
> Some of the evidence layers likely capture overlapping properties.

For example, annotation completeness and literature evidence may both strongly favor well-studied genes. Localization evidence may also partly reflect the same published experiments used in ciliary databases or literature mining.

This could help explain why highly characterized housekeeping genes receive unexpectedly high scores.

I therefore think an inter-layer correlation analysis would be useful. A leave-one-layer-out analysis would also help establish whether the integrated model provides information beyond the strongest individual components.

In addition, the Methods could provide a little more detail about how multiple sources are combined within each layer, particularly for expression, localization, annotation, and literature evidence.

Response. We added pairwise observed-gene correlations with sample sizes, all six leave-one-layer-out rankings and matched-pair omissions. Annotation–literature is the strongest correlation (ρ=0.3704; n=19,114); moderate correlations are not evidence of independent biology. Animal/literature omissions weaken matched separation, especially jointly. Methods and S3 now describe expression source/ordinal semantics and available-component normalization, annotation missing-subcomponent zero contribution, localization HPA reliability/embedded-list handling, animal term union and literature tier precedence. We corrected descriptions that conflated Approved/Uncertain antibody evidence with computational predictions or embedded lists with newly acquired proteomics datasets. We now disclose that 26/28 SYSCILIA controls overlap the undated embedded list, which enters both raw localization and the HIGH gate; exact literals and alias overlaps are supplied. Sensory literature queries include Usher syndrome terms. Shared underlying studies and disease knowledge therefore contribute to multiple layers; separate computation does not mean independent evidence.

Changes/evidence. Main Methods “Evidence layers”; Results missingness/sensitivity/comparator sections; S3–S5; results/layer_correlations.tsv, baseline_comparison.tsv; phase4_results/fixed_pair_sensitivities.tsv.

Revised PDF locations. Manuscript p. 9–10 (Evidence layers); Manuscript p. 23–24 (Sensitivity analysis); Supplementary Methods p. 2–3 (S3. Implemented); Supplementary Methods p. 4 (S4. Control traces); Supplementary Methods p. 4–5 (S5. Frozen).

R1-major-8. Seed-confounded mantis-ml benchmark; temporal or simple baselines

Comment.

The mantis-ml comparison is informative but does not provide a strong external benchmark

The manuscript appropriately acknowledges that the mantis-ml comparison is difficult to interpret because 33 of the 35 shared positive controls were used as mantis-ml training seeds.

For that reason, the higher control recovery of mantis-ml cannot really be interpreted as a conventional head-to-head benchmark. Conversely, the comparison also does not establish how well UsherPipe performs on genuinely novel genes.

If possible, a temporal validation would be particularly interesting—for example, testing whether the pipeline can prioritize genes that were associated with the relevant biology only after a defined historical cutoff.

If this is not practical, simpler baselines would still be valuable. The authors could compare the full six-layer score against each major layer alone and against an equal-weight model. This would help demonstrate whether the integrated framework provides a meaningful benefit over simpler scoring strategies.

Response. The cached-output comparison retains the 33/35 shared-positive seed overlap and only two non-seeds, without claiming head-to-head novel-gene performance or rerunning mantis-ml training. Simple

baselines are now reported: animal-only is strongest for Usher and literature-only for SYSCILIA control recovery; both exceed the default score for their corresponding group.

We subsequently completed the requested temporal assessment, extending the preserved restricted pilot. RetNet/PanelApp catalogs, gene/alias searches and primary reports were used to review dates, unrelated target families, inheritance and functional/replication evidence. The 45 principal cases, sources, code and comparisons were fixed before expanded scores; development controls and the three previously inspected pilot examples were excluded. We distinguish cases with no earlier human disease report located in the adopted audit (A) from first target associations after another phenotype (B); no universal first-ever claim is made. The adopted cohort is not exhaustive, and 895 unconfirmed/unnominated catalog records remain uncertain rather than clinical negatives.

Using a 31 December 2020 evidence cutoff, we reconstructed gnomAD v2.1.1, HPA v20, GTEx v8, GO, original 2019 retinal matrices, native MGI/IMPC/ZFIN evidence and archived gene2pubmed. Two co-primary versions omit or include date-bounded literature queried from modern text; 21 additional comparisons were specified before outcomes. Among 41 in-universe cases, top 100/HIGH recovery was 0/41 in both versions, top 1,000 was 2/41 and 3/41, and top 10% was 5/41 and 6/41. Four noncoding cases remained nonrecoveries in the 45-case denominator. These negative/weak results are in the Abstract, Results, Table 4 and Figure 8. Genome-wide HIGH contains only zero/five genes. Category A top-1,000 recovery is 1/36 and 2/36; category B is 1/5 in both. Recovery is descriptively near random ordering rather than demonstrated enrichment. No principal case is a new Usher association. These are the preserved outcomes of the original ascertainment; final clinical eligibility remains provisional where concerns raised during subsequent author review are unresolved. We did not replace these outcomes with favorable sensitivity results.

Additional human author review. A human author reviewed primary reports for all 45 cases, examining reported families/pedigrees, phenotypes, inheritance, functional mechanisms and publication dates. This review took place after the frozen temporal analysis. It identified two issues relevant to interpretation: the diversity of disease mechanisms and the strength of family evidence.

The family-evidence concerns involved **PDIA6, VWA8 and PLCG1**. The reviewed initial PDIA6 report described one affected individual from one family. The earlier VWA8 developmental-disorder report described nine affected individuals from one family, whereas the temporal target was a later retinal association. For PLCG1, the review distinguished a single-family immune-disorder report from a later multisystem report with auditory/ocular involvement and four families. These findings show why family support must be evaluated for the specific gene–target-phenotype association across relevant reports. The frozen screening ledger also cites later PDIA6 reports (PMIDs 35856135, 40974269), recording three families overall and a conservative lower bound of two target families; the VWA8 retinal reports (PMIDs 37012052, 40638000) give a target-family lower bound of two, and PLCG1 has a conservative target-family lower bound of three. These are the retained screening classifications, not new human inclusion decisions. The index-paper concerns do not by themselves constitute three exclusions. Eligibility of the flagged cases remains under review; the original temporal results are retained.

Representative biological distinctions are as follows:

- **CRLS1:** the reported disease is mitochondrial, involving cardiolipin deficiency, with retinal and auditory manifestations (Lee et al., 2022; PMID 35147173).
- **CTNND1:** the reported retinal association is familial exudative vitreoretinopathy through the Wnt/cadherin axis (Yang et al., 2022; PMID 35700046).
- **DHRSX:** the reported disease is a congenital disorder of glycosylation arising from impaired dolichol biosynthesis; DHRSX is an enzyme in this pathway (Wilson et al., 2024; PMID 38821050).

- **RNU4-2 and the RNU6 genes:** the relevant reports concern spliceosomal small nuclear RNAs and nonsyndromic retinitis pigmentosa (Quinodoz et al., 2026; PMID 41513982).

A post-hoc contemporary GO classification further describes the same 45 cases (S8; Additional file 13, Figure S1). BP/MF/CC coverage was 41/42/44 cases; overlapping categories included cilium and mitochondrion (eight each), cilium organization (four) and mRNA metabolic process (five). None of the 11,388 primary or 10,459/2,750 sensitivity GO tests passed pooled BH FDR < 0.05. This functional description does not alter the historical scores or supply independent predictive validation.

These examples clarify the breadth of the prespecified sensory-disease cohort. Retinal or auditory involvement alone does not establish a ciliopathy, so aggregate recovery in this cohort should be interpreted within that broader scope.

The publication-date audit distinguished first public target reports, including preprints and early-online versions, from submission, acceptance and later issue dates. Most date differences reflected these publication stages. The timing criterion is a supported first-target publicity interval wholly after 31 December 2020; an exact first-public day is unnecessary when that interval is unambiguously post-cutoff. The reviewed sources supported this temporal classification, with publication-version details and date intervals retained in the audit.

This is a retrospective evaluation of historical analogues, not an exact production replay or prospectively developed predictor. Historical source substitutions, missing HCOP/annotation channels, modern-text uncertainty and conditional catalog ascertainment are explicit. The temporal study does not include a harmonized mantis-ml rerun or validate its comparative novel-gene performance. The preserved pilot results remain separately documented.

Changes/evidence. Methods temporal design; Results temporal assessment, sensitivity and mantis-ml sections; Discussion/Conclusions; Table 4/Figure 8; S4/S6; Additional files 11/12 with protocols, clinical ledger, baselines and full outputs.

Revised PDF locations. Manuscript p. 23–24 (Sensitivity analysis); Manuscript p. 24–26 (Retrospective temporal evaluation); Manuscript p. 26–27 (Biological scope of the temporal cohort); Manuscript p. 27–28 (Comparison with mantis-ml); Supplementary Methods p. 4 (S4. Control traces); Supplementary Methods p. 5–11 (S6. Retrospective); Supplementary Methods p. 11–13 (S8. Contemporary).

R1-major-9. Versions, timestamps, locked environment, commands and actual supplementary checksums

Comment.

Reproducibility should be strengthened

The authors have made a serious effort to document reproducibility, which is appropriate for a BMC Bioinformatics manuscript. However, the current provenance information is still incomplete.

The reproducibility manifest records only a subset of source URLs and versions and does not include retrieval timestamps for the recorded inputs.

For a computational pipeline paper, I would strongly encourage the authors to provide, where possible:

exact database versions or snapshot identifiers;

source URLs or accession numbers;

retrieval dates;

checksums for local input files;

a locked software environment or dependency file;

the commands needed to reproduce the figures and tables;

ideally, a containerized environment or an equivalent reproducible setup.

I also recommend checking the consistency of the submitted supplementary files against the checksum manifest. In the material provided for review, the checksum associated with the expression-shortlist report does not appear to match the submitted Additional file 6. This may simply reflect a conversion from a repository Markdown file to the journal TXT file, but if so, that distinction should be documented.

Response. We supply four schema-checked canonical derived-input tables, preserved baseline prose, explicit database/cache/output options, frozen ClinGen/HGNC exports and exact tested dependency locks. Production numerical replay is distinguished from complete raw-source acquisition; restored donor-derived annotation/animal fields and originally unknown retrieval dates remain disclosed. The earlier clean Python 3.13.1/macOS-arm64 replay reproduced 28 TSVs, including eligibility/matching. The expanded temporal run verified 24 numerical source files (204,322,926 bytes) and a committed dependency/source/clinical freeze; a separate clean reconstruction reproduced all nine temporal TSVs and the manifest byte-for-byte. The current full suite, including eight GO tests, passed 362 tests with 17 existing warnings. Source versions, URLs, accessions, retrieval metadata where recorded, local SHA-256 and figure/table commands are included in Additional files 11/12 and S7. Linux/container parity is not claimed.

We confirmed the original mismatch: repository expression-shortlist Markdown hashes to e046dbcd..., while the submitted Additional file 6 TXT hashes to 13fc6ddd.... The revised Additional file 10 instead hashes the actual delivered additional files, including freshly converted files 3/6; it excludes itself rather than claiming a self-hash. A separate package manifest records the delivered manuscript/response DOCX and PDF bytes, source hashes and source-analysis checkpoints. Both revised TXT files are generated from their named Markdown sources. Rendered document checks and a page location index accompany this revision. Large raw archives/clinical caches are not silently represented as distributed or backed up.

Changes/evidence. Methods “Reproducibility”; S7; Additional files 4/10/11/12; package manifest and response location index. The complete executable numerical replay remains in the repository.

Revised PDF locations. Manuscript p. 15 (Reproducibility); Manuscript p. 33–34 (Availability of data and materials); Manuscript p. 34–36 (Additional files); Supplementary Methods p. 11 (S7. Numerical replay).

R1-major-10. Retina/proxy expression, absence of hair-cell production data and proxy sensitivity

Comment.

The production expression layer seems more retina-oriented than fully Usher-specific

The manuscript appropriately emphasizes that Usher syndrome involves both retinal and auditory biology. However, the production expression layer does not appear to include cochlear hair-cell data. The GSE135913 cochlear analysis is instead treated as a post-hoc exploratory analysis based on two accepted fetal samples.

At the same time, cerebellum is retained as a proxy tissue in the production score.

I think the authors should explain this choice more clearly, because tissue specificity is presented as an important feature distinguishing UsherPipe from more general gene-prioritization tools.

It would be useful to show how much the expression layer contributes to the ranking of established Usher genes, and whether removing the proxy tissue materially changes the results.

Until direct auditory-cell data are incorporated into the production score, the term “Usher-specific tissue weighting” may be somewhat stronger than the available data support.

Response. We replace “Usher-specific tissue weighting” with explicit retina/proxy expression terminology. Cerebellum is an adult neural proxy, not auditory evidence. Production lacks direct cochlear hair-cell measurements; developmental or auditory-restricted genes can be disadvantaged when other evidence is also limited. The fetal GSE135913 analyses remain exploratory views within an already selected HIGH tier and do not repair this production gap.

Expression contributes 15.5–27.5% of individual Usher controls’ weighted support. Removing cerebellum and recomputing source transforms yields whole-ranking ρ=0.9496 and median absolute shift 4.92 percentile points. HIGH is 60 versus 62, with only 37 shared genes (25 leave/23 enter): USH2A/MYO7A enter, CDH23 leaves and PCDH15 remains. We report this material dependence without selecting a revised score using control recovery. Source coverage and background-only Tau contributions are made explicit.

Changes/evidence. Methods expression layer/Table 1; Results “Expression dependence”; Discussion; S3; results/usher_expression_diagnostics.tsv, baseline_comparison.tsv, per_gene_diagnostics.tsv.

Revised PDF locations. Manuscript p. 24 (Expression dependence); Manuscript p. 29–31 (Limitations); Supplementary Methods p. 1–2 (S1. Optional); Supplementary Methods p. 2–3 (S3. Implemented).

Reviewer 1 — minor comments

R1-minor-1. Define seed-free at first introduction

Comment.

> The term “seed-free” should be defined clearly when it is first introduced. Known genes are not used in the raw composite score, but control behavior did influence the downstream cilia-signal gate.

Response. Abstract Results now defines raw scoring as using neither known disease training labels nor similarity seeds and immediately distinguishes the control-informed downstream gate. The fuller Methods definition explicitly permits known-gene annotations and disease-name literature terms: 26/28 SYSCILIA controls overlap the embedded scoring/gating compendium. Seed-free is restricted to absence of training labels and similarity seeds, not absence of known-gene knowledge.

Revised PDF locations. Manuscript p. 1–2 (Abstract); Manuscript p. 11–12 (NULL-aware composite scoring); Supplementary Methods p. 2–3 (S3. Implemented).

R1-minor-2. Use priority tiers rather than confidence tiers

Comment.

> I would consider replacing “confidence tier” with “priority tier.” HIGH, MEDIUM, and LOW are rule-based prioritization categories rather than calibrated statistical confidence levels.

Response. Revised manuscript and supplementary prose use “priority tier”, including figure captions. The legacy data column confidence_tier remains for schema compatibility, explicitly documented as a rule-based priority category rather than calibrated statistical confidence. Original frozen artifacts retain their historical wording.

Revised PDF locations. Manuscript p. 11–12 (NULL-aware composite scoring); Manuscript p. 19–20 (Internal control recovery); Supplementary Methods p. 2–3 (S3. Implemented).

R1-minor-3. Check canonical/HGNC terminology

Comment.

> Gene-identifier terminology should be checked carefully. The manuscript acknowledges that many displayed gene names are not formally validated current HGNC symbols, so wording such as "canonical identifier" should only be used where it is strictly accurate.

Response. We describe retained analysis IDs/labels, duplicate-symbol consolidation and 573 ENSG fallback labels without calling them validated current HGNC symbols. Additional file 1's description is corrected to retained analysis identifiers. Current HGNC mappings are used for comparator eligibility and the separate contemporary GO classification; these do not retrospectively validate all production names.

Revised PDF locations. Manuscript p. 8–9 (Gene universe); Supplementary Methods p. 1–2 (S1. Optional).

R1-minor-4. Main-text control tier distribution

Comment.

> The distribution of established Usher and SYSCILIA controls across the final tiers should be included in the main Results section.

Response. Added 2 HIGH/7 MEDIUM Usher and 1 HIGH/27 MEDIUM SYSCILIA in Results, with zero LOW/excluded in both groups and all nine Usher traces in Table 3 (R1-major-3).

Revised PDF locations. Manuscript p. 19–20 (Internal control recovery); Supplementary Methods p. 4 (S4. Control traces).

R1-minor-5. Background sampling and random seed

Comment.

> Please state how the random background genes shown in the control-recovery analysis were sampled and provide the random seed.

Response. Figure 5 caption and S7 specify 500 non-control scored labels sampled without replacement, NumPy default_rng(42). The revision figure code orders the 20,016 eligible background labels by retained Ensembl ID and exports sampled IDs and percentiles, making selection reproducible rather than dependent on database row order.

Revised PDF locations. Manuscript p. 40 (Figure 5. Internal control recovery); Supplementary Methods p. 11 (S7. Numerical replay).

R1-minor-6. Permanent exact-version archive/DOI if possible

Comment.

> In addition to the GitHub commit, I would encourage archiving the exact software version used for the manuscript in a permanent repository with a DOI if possible.

Response. We retain the originally cited public code/data commit 4cf8c5b, rechecked directly on GitHub, and separately identify the preserved original submitted files: the package was prepared at f889ec3, and ed2d00d also preserves the August technical-check amendment. We identify exact revision/temporal analysis checkpoints, supply source/input/output hashes, dependency locks and numerical replay bundles, and hash the actual delivered files. No permanent DOI-bearing archive has been minted, and we do not imply that Git is equivalent to one. The reproducibility bundles and exact Git versions provide a version-specific record; a DOI deposit has not been made because the original raw-source acquisition is incomplete and final clinical eligibility remains provisional where author-review concerns are unresolved. We prefer to finalize that scope before describing a permanent deposited version. This does not prevent inspection of the exact delivered data or current Git checkpoints.

Changes/evidence. Data availability; S7; Additional files 10-12; exact temporal outcome/audit checkpoint 736e16f and production revision checkpoint 0e1634a.

Revised PDF locations. Manuscript p. 33–34 (Availability of data and materials); Supplementary Methods p. 11 (S7. Numerical replay).

R1-minor-7. Consider shortening biological background

Comment.

> The biological background is thorough, but somewhat long for a BMC Bioinformatics paper. Some of the detailed discussion of Usher molecular biology could be shortened so that more emphasis can be placed on the computational design, validation strategy, benchmarking, and scoring assumptions.

Response. We retain the mechanistic rationale needed to understand sensory/ciliary scope and the distinction between actin-based stereocilia and true cilia. We clarify the scope paragraph and add the requested references, while expanding computational design, diagnostics and limitations. The mechanistic background explains why the biological target is broader than an Usher diagnosis, while the added computational design and validation sections address the methodological emphasis requested. Rendered layout/readability was checked in the revised document package.

Revised PDF locations. Manuscript p. 3–4 (Usher syndrome as a specialized sensory ciliopathy); Manuscript p. 4–5 (Ciliary and stereociliary biology underlying Usher syndrome).

Reviewer 2

Comment.

> The figures were not in an accessible form. So I could not review this.

Response. Thank you for identifying the access problem. The prepared revision material includes the figures embedded with their legends in the manuscript and as separate figure files. Before resubmission, we will check that the journal-generated review PDF renders each figure and that the uploaded figure files are available to reviewers. We would welcome the reviewer's assessment once the figures are accessible.

Reviewer 3

R3-1. Add support for the ciliary/periciliary Background sentence

Comment.

> In Background, in the sentence beginning "Because the known Usher genes encode ciliary and periciliary proteins acting at hair-cell stereocilia, the photoreceptor connecting cilium, and the periciliary membrane complex, Usher syndrome may…", a supporting reference should be added.

Response. The revised sentence describes several Usher proteins in stereociliary and photoreceptor connecting-cilium/periciliary complexes, supported by references [8, 9, 10, 11]. This avoids treating every Usher protein as exclusively ciliary and explicitly distinguishes actin-based stereocilia in the next paragraph. See Background "Usher syndrome as a specialized sensory ciliopathy".

Revised PDF locations. Manuscript p. 3–4 (Usher syndrome as a specialized sensory ciliopathy).

R3-2. Seed-free versus post-hoc gate calibration and HIGH interpretation

Comment.

> The manuscript describes UsherPipe as a seed-free approach, but the cilia-signal gate was defined after examining control and negative-control behavior. The authors note that this is a post-hoc

calibration. It would be helpful to clarify how this affects the interpretation of control-recovery performance and the definition of the HIGH tier.

Response. We define seed-free narrowly as no training labels or similarity seeds, disclose embedded known-gene/disease-term inputs, and separate raw scoring from downstream control-informed calibration in the Abstract, Methods and Results. HIGH is a heuristic priority tier, not validated specificity or disease probability. Table 3 reports the control filtering losses directly, and matched comparator results cannot undo reused-positive calibration. See R1-major-2/3 and Supplementary Methods S3/S5/S6.

Revised PDF locations. Manuscript p. 11–12 (NULL-aware composite scoring); Manuscript p. 19–20 (Internal control recovery); Supplementary Methods p. 2–3 (S3. Implemented); Supplementary Methods p. 4–5 (S5. Frozen); Supplementary Methods p. 5–11 (S6. Retrospective).

R3-3. Justify final weights and discuss logistic-ranking sensitivity

Comment.

The manuscript reports that alternative weighting strategies, including logistic-regression-based weighting, substantially changed the genome-wide ranking (default vs. logistic regression rank ρ = 0.646). These alternatives were not adopted due to concerns about control reuse and over-specialization. Additional justification for the final predefined weights would be useful, along with a discussion of how this degree of ranking sensitivity affects the robustness of candidate prioritization.

Response. We explain the limited design rationale for 0.20/0.20/0.15/0.15/0.15/0.15 and explicitly label exact values heuristic. Equal, individual-layer and leave-one-layer-out baselines and top25/50/100 persistence show that high global correlation does not ensure stable shortlists. Only 41 default top 100 genes persist in all 25 modest configurations; animal-only better recovers Usher controls and literature-only better recovers SYSCILIA controls (4/9 and 21/28 at top 100, respectively). Logistic rank ρ=0.646 remains a material alternative, rejected as production fitting because of control reuse/over-specialization, not because it is numerically equivalent. See R1-major-5 and Results/S2/S4.

Revised PDF locations. Manuscript p. 23–24 (Sensitivity analysis); Supplementary Methods p. 2 (S2. Data-driven); Supplementary Methods p. 4 (S4. Control traces).

R3-4. Independent support for genuinely novel disease-gene prioritization

Comment.

The recovery of known Usher and ciliary genes shows that the pipeline captures relevant disease biology. However, because the cilia-signal gate was calibrated using these same controls, it would be helpful to clarify what independent evidence supports UsherPipe's ability to prioritize genuinely novel disease genes rather than mainly rediscovering genes with established Usher/ciliary characteristics.

Response. We added a retrospective, temporally separated later-association evaluation and report its weak results rather than relying solely on known-control recovery. The 45 cases were fixed from clinical/date evidence before expanded ranking; development controls and the previously inspected pilot cases were excluded. Forty-one were in the historical protein-coding universe, with four noncoding cases retained as end-to-end failures. Neither five-layer nor six-layer analogue recovered a principal case at top 100 or HIGH; top 1,000 recovery was 2/41 and 3/41. Median ascending percentiles were 46.23/44.95. Genome-wide HIGH contains only zero/five genes. Top-1,000 recovery is descriptively close to random-ordering expectation of approximately 2.14 cases (not a formal null test). Category A top-1,000 recovery was 1/36 and 2/36; category B was 1/5 in both. No principal case represents a new Usher gene. No principal case reached HIGH under any of 23 schemes.

This is evidence about the limitations of later-association recovery under the stated historical conditions, not evidence of reliable novel Usher-gene discovery. The initial clinical/date ascertainment was AI-assisted. Subsequent human author review covered primary reports for all 45 cases, including family/pedigree evidence, phenotypes, functional mechanisms and publication chronology. It highlighted mitochondrial disease (CRLS1), retinal vascular disease (CTNND1), glycosylation disease (DHRSX) and spliceosome-related retinal disease (RNU genes), together with family-evidence concerns involving PDIA6, VWA8 and PLCG1. R1-major-8 describes these findings and their implications for case eligibility. The temporal cohort consequently represents the prespecified broader sensory/ciliopathy scope.

The score/gate was developed after the historical cutoff, several source channels differ from production, modern text retains uncertainty, and catalog ascertainment is non-exhaustive. We therefore retain a hypothesis-generation interpretation throughout Abstract, Discussion and Conclusions. The matched comparator analysis and control recovery also remain descriptive; neither validates clinical specificity or HIGH disease probability.

Changes/evidence. Methods/Results temporal sections; Table 4/Figure 8; S6; Additional file 12 with the 45-case roster, date/family evidence, exclusions, source contract, all outcomes and reconstruction code.

Revised PDF locations. Manuscript p. 12–14 (Retrospective evaluation of historical analogues); Manuscript p. 24–26 (Retrospective temporal evaluation); Manuscript p. 26–27 (Biological scope of the temporal cohort); Manuscript p. 29–31 (Limitations); Supplementary Methods p. 5–11 (S6. Retrospective); Supplementary Methods p. 11–13 (S8. Contemporary).

R3-5. Impact of excluding hair-cell expression and developmental bias

Comment.

> Because Usher syndrome affects cochlear hair cells, the authors could more directly discuss the impact of excluding hair-cell expression data from the production score. The exploratory fetal cochlear analyses are useful, but they do not contribute to the final composite score. It would be helpful to comment on whether the absence of suitable hair-cell data may disadvantage genes with more limited or developmentally restricted evidence.

Response. Main Methods/Results/Discussion now state the lack of production auditory-cell evidence and possible disadvantage for auditory-restricted or developmentally restricted genes. The fetal analyses do not enter the score, are based on two marker-selected samples and already selected HIGH genes, and do not establish adult expression or independent validity. We quantify expression contribution and substantial cerebellum-proxy sensitivity rather than implying complete Usher-tissue coverage. See R1-major-10, S1/S3.

Revised PDF locations. Manuscript p. 24 (Expression dependence); Manuscript p. 29–31 (Limitations); Supplementary Methods p. 1–2 (S1. Optional); Supplementary Methods p. 2–3 (S3. Implemented).

R3-6. General gene importance/research intensity versus disease specificity

Comment.

> A broader conceptual issue concerns the biological specificity of the composite score. Several evidence layers, such as gnomAD constraint, functional annotation completeness, and publication-based evidence, may reflect general gene importance or research intensity rather than disease-specific involvement in Usher syndrome. This is particularly relevant because housekeeping genes ranked very highly in the raw multi-evidence score (median at the 94.8th percentile). Clarifying how the composite score distinguishes general biological importance from evidence specifically supporting a role in Usher syndrome or related ciliopathies would strengthen the interpretation. The authors might also consider discussing whether additional disease-specific negative controls or specificity analyses could help quantify this distinction.

Response. We retain the 94.8th-percentile housekeeping failure and clarify that constraint, annotation depth and publication context can encode general importance or research effort. We added publication/GO-matched unrelated-disease comparators, correlations, source-completeness balance and common-layer/omission analyses. Separation remains with common layers but weakens to 55.6%/57.1% after animal/literature omission; clinical-label exclusions also overlap features relevant to the score. Literature-only medians reach 99.13%/99.78% for known controls and its top 100 contains 2/9 Usher and 21/28 SYSCILIA controls, emphasizing known biology and research/curation rather than independent discovery. These checks do not make the composite Usher-specific or remove all ascertainment. Rigorously adjudicated clinical negatives remain unavailable. The separate later-association cohort has now been evaluated, with weak outcomes that do not establish disease-specific predictive validity. See R1-major-4/7 and S3–S5.

Human author review further clarifies the cohort's biological scope. The mitochondrial, retinal vascular, glycosylation and RNA-splicing examples provide concrete reasons to distinguish broader sensory-disease prioritization from ciliopathy-specific performance (R1-major-8). The added contemporary GO classification is consistent with functional heterogeneity, but no enrichment passed FDR < 0.05; we do not treat it as evidence of cilia specificity (S8; Additional file 13).

Revised PDF locations. Manuscript p. 20–21 (Negative controls); Manuscript p. 21–22 (Matched unrelated-disease comparators); Manuscript p. 23–24 (Sensitivity analysis); Manuscript p. 26–27 (Biological scope of the temporal cohort); Supplementary Methods p. 2–3 (S3. Implemented); Supplementary Methods p. 4 (S4. Control traces); Supplementary Methods p. 4–5 (S5. Frozen); Supplementary Methods p. 11–13 (S8. Contemporary).